

Ch 9: Light - Reflection and Refraction

Question Bank | 2M | 3M | 5M

SECTION A - 2 MARKS

- Q1. What is light? Why is light considered a form of energy? State one property that makes light useful for seeing objects. (PYQ 2019)
- Q2. Define reflection of light. State the two laws of reflection. (PYQ 2020)
- Q3. What is a plane mirror? Mention two characteristics of the image formed by a plane mirror. (PYQ 2022)
- Q4. Define the terms: (i) incident ray (ii) reflected ray (iii) normal (iv) angle of incidence. (PYQ 2019)
- Q5. What is lateral inversion? Give one daily-life example where lateral inversion is observed. (PYQ 2020)
- Q6. Define spherical mirror. Name the two types of spherical mirrors with one use of each. (PYQ 2022)
- Q7. Differentiate between concave mirror and convex mirror on the basis of reflecting surface and nature of focus. (PYQ 2024)
- Q8. Define pole, centre of curvature, radius of curvature and principal axis of a spherical mirror. (PYQ 2019)
- Q9. What is principal focus of a concave mirror? Why is the focus of a convex mirror called virtual? (PYQ 2020)
- Q10. Write the relation between focal length and radius of curvature of a spherical mirror. If $R = 30\text{ cm}$, find f . (PYQ 2022)
- Q11. State the sign convention used for spherical mirrors. What sign is given to object distance and focal length of a concave mirror? (PYQ 2019)
- Q12. Write the mirror formula. Name the quantities represented by u , v and f . (PYQ 2020)
- Q13. Define magnification produced by a spherical mirror. Write its formula in terms of image distance and object distance. (PYQ 2022)
- Q14. What does a positive magnification indicate? What does a negative magnification indicate? (PYQ 2019)
- Q15. An object is placed at the centre of curvature of a concave mirror. State the position, size and nature of the image. (PYQ 2020)
- Q16. An object is placed between pole and focus of a concave mirror. State the nature and position of the image formed. (PYQ 2024)
- Q17. Why is a convex mirror used as a rear-view mirror in vehicles? Give two reasons. (PYQ 2019)
- Q18. Why are concave mirrors used in torches and vehicle headlights? (PYQ 2022)
- Q19. Define refraction of light. Why does refraction occur when light passes from one medium to another? (PYQ 2020)
- Q20. State the laws of refraction of light. (PYQ 2019)
- Q21. Define refractive index of a medium. Write its formula in terms of speed of light. (PYQ 2022)
- Q22. The refractive index of glass is 1.5. What does this statement mean? (PYQ 2020)

- Q23. What happens to a ray of light when it travels from air to glass? Does it bend towards or away from the normal? (PYQ 2019)
- Q24. What happens to a ray of light when it travels from glass to air? State the direction of bending. (PYQ 2022)
- Q25. Why does a pencil appear bent when partly immersed in water? (PYQ 2020)
- Q26. Why does a coin placed in water appear raised? Name the phenomenon involved. (PYQ 2019)
- Q27. Define lateral displacement through a rectangular glass slab. On what factors does it depend? (PYQ 2024)
- Q28. What is a lens? Name the two main types of spherical lenses. (PYQ 2022)
- Q29. Differentiate between convex lens and concave lens based on thickness and effect on parallel rays. (PYQ 2020)
- Q30. Define optical centre and principal axis of a lens. (PYQ 2019)
- Q31. What is principal focus of a convex lens? What is principal focus of a concave lens? (PYQ 2022)
- Q32. Write the lens formula. State the sign of focal length for convex and concave lenses. (PYQ 2020)
- Q33. Define power of a lens. Write its SI unit. (PYQ 2019)
- Q34. What is one dioptre? Find the power of a lens of focal length 50 cm. (PYQ 2022)
- Q35. A lens has power -2 D. Identify the type of lens and find its focal length. (PYQ 2024)
- Q36. What is the nature of image formed by a convex lens when the object is between F_1 and $2F_1$? (PYQ 2020)
- Q37. What is the nature of image formed by a concave lens for any position of object? (PYQ 2019)
- Q38. Why is a convex lens used as a magnifying glass? Where should the object be placed? (PYQ 2022)
- Q39. State two differences between real image and virtual image. (PYQ 2020)
- Q40. Why should ray diagrams be drawn using at least two standard rays? (PYQ 2024)

SECTION B - 3 MARKS

Q1. Explain reflection of light from a plane mirror. State the laws of reflection and prove that the image formed by a plane mirror is virtual and erect. (PYQ 2019)

Q2. Compare concave and convex mirrors:

Basis	Concave Mirror	Convex Mirror
Reflecting surface	Inner curved surface	Outer curved surface
Nature of focus	Real focus	Virtual focus
Image type	Real or virtual depending on object position	Always virtual, erect and diminished
Common use	Shaving mirror, torch reflector	Rear-view mirror

Q3. Draw ray diagrams for image formation by a concave mirror when the object is: (i) at infinity (ii) beyond C (iii) at C. State position, size and nature of image. (PYQ 2022)

Q4. Draw ray diagrams for image formation by a concave mirror when the object is: (i) between C and F (ii) at F (iii) between F and P. State the nature of image in each case. (PYQ 2020)

Q5. Explain the uses of concave mirrors in: (i) shaving mirror (ii) headlights (iii) solar furnace. Give reason for each use. (PYQ 2019)

Q6. Explain the uses of convex mirrors as rear-view mirrors and security mirrors. Mention the nature of image formed. (PYQ 2022)

Q7. Derive or explain the mirror formula using sign convention. Mention the meaning of each term and the importance of signs. (PYQ 2020)

Q8. A concave mirror has focal length 15 cm. An object is placed 30 cm in front of it. Find the image distance and nature of image. (PYQ 2019)

Q9. An object of height 4 cm is placed 20 cm in front of a concave mirror of focal length 10 cm. Find the magnification and image height. (PYQ 2024)

Q10. Compare real and virtual images:

Basis	Real Image	Virtual Image
Formation	Actual meeting of rays	Apparent meeting of rays
Screen	Can be obtained on screen	Cannot be obtained on screen
Orientation	Generally inverted	Generally erect
Example	Image by concave mirror beyond F	Image by plane mirror

Q11. Explain refraction through a rectangular glass slab with a labelled ray diagram. Why is the emergent ray parallel to the incident ray? (PYQ 2022)

Q12. State the laws of refraction. Explain why light bends towards the normal when it travels from rarer to denser medium. (PYQ 2020)

Q13. Compare reflection and refraction:

Basis	Reflection	Refraction
Meaning	Bouncing back of light in same medium	Bending of light on passing to another medium
Medium	Same medium	Two transparent media
Speed change	No change if medium remains same	Speed changes
Law used	Laws of reflection	Laws of refraction / Snell law

Q14. Explain refractive index. If speed of light in air is 3×10^8 m/s and speed in glass is 2×10^8 m/s, calculate refractive index of glass. (PYQ 2019)

Q15. Why does a swimming pool appear shallower than it actually is? Explain using refraction. (PYQ 2022)

Q16. Explain the refraction of light through a glass slab and define lateral displacement. State factors affecting lateral displacement. (PYQ 2024)

Q17. Compare convex lens and concave lens:

Basis	Convex Lens	Concave Lens
Thickness	Thicker at centre	Thinner at centre
Action on rays	Converges parallel rays	Diverges parallel rays
Focus	Real focus	Virtual focus
Focal length sign	Positive	Negative
Image nature	Real/virtual depending on object position	Always virtual, erect and diminished

Q18. Draw ray diagrams for a convex lens when object is: (i) at infinity (ii) beyond $2F_1$ (iii) at $2F_1$. State image position, size and nature. (PYQ 2020)

Q19. Draw ray diagrams for a convex lens when object is: (i) between F_1 and $2F_1$ (ii) at F_1 (iii) between F_1 and optical centre. (PYQ 2019)

Q20. Draw a ray diagram for image formation by a concave lens. Why is the image always virtual, erect and diminished? (PYQ 2022)

Q21. Explain the sign convention for lenses. Give signs of u , v , f and h for real and virtual images. (PYQ 2020)

Q22. Use lens formula to solve: An object is placed 30 cm from a convex lens of focal length 20 cm. Find image distance and nature of image. (PYQ 2019)

Q23. A concave lens of focal length 25 cm forms an image 15 cm from the lens on the same side as object. Find object distance. (PYQ 2024)

Q24. Define magnification of a lens. A convex lens forms an image twice the size of the object. What does the sign of magnification tell about image nature? (PYQ 2022)

Q25. Explain power of a lens. Calculate the power of: (i) convex lens of focal length 25 cm (ii) concave lens of focal length 50 cm. (PYQ 2020)

Q26. Compare mirror formula and lens formula:

Basis	Mirror Formula	Lens Formula
Formula	$1/f = 1/v + 1/u$	$1/f = 1/v - 1/u$
Applies to	Spherical mirrors	Spherical lenses
Main distances	Measured from pole	Measured from optical centre
Sign of f	Concave negative, convex positive	Convex positive, concave negative

Q27. Why is a concave lens called a diverging lens and a convex lens called a converging lens? Explain with ray behaviour. (PYQ 2019)

Q28. What happens when a ray of light passes through: (i) optical centre of lens (ii) focus of convex lens (iii) parallel to principal axis? (PYQ 2022)

Q29. Explain the relation between object position and magnification for a convex lens. When is enlarged real image formed? When is enlarged virtual image formed? (PYQ 2020)

Q30. A lens has power +4 D. Find its focal length and identify whether it is converging or diverging. (PYQ 2019)

Q31. A student obtains a blurred image on screen using a convex lens. State three possible adjustments to get a sharp image. (PYQ 2024)

Q32. Write differences between spherical mirror and spherical lens:

Basis	Spherical Mirror	Spherical Lens
Surface	Reflecting curved surface	Transparent refracting surfaces
Phenomenon	Reflection	Refraction
Reference point	Pole	Optical centre
Formula	Mirror formula	Lens formula

Q33. Explain four precautions to be followed while drawing ray diagrams in geometrical optics. (PYQ 2022)

Q34. Solve a numerical using mirror formula: $u = -20$ cm, $f = -10$ cm. Find v , magnification and nature of image. (PYQ 2020)

Q35. Solve a numerical using lens formula: $u = -15$ cm, $f = +10$ cm. Find v , magnification and nature of image. (PYQ 2019)

SECTION C - 5 MARKS

Q1. Explain reflection of light in detail. State the laws of reflection. Draw a neat labelled diagram showing incident ray, reflected ray, normal, angle of incidence and angle of reflection. Explain image formation by a plane mirror. (PYQ 2022)

Q2. Describe concave and convex mirrors. Define pole, centre of curvature, radius of curvature, principal axis, aperture, principal focus and focal length. Write the relation between radius of curvature and focal length. (PYQ 2020)

Q3. Draw ray diagrams for image formation by a concave mirror for all major object positions: at infinity, beyond C, at C, between C and F, at F, and between F and P. Write position, size and nature of image in each case. (PYQ 2019)

Q4. Draw ray diagrams for image formation by a convex mirror. Explain why the image formed by a convex mirror is always virtual, erect and diminished. Give two practical uses. (PYQ 2022)

Q5. Write a complete note on sign convention for spherical mirrors. Explain how object distance, image distance, focal length and height are assigned positive or negative signs. Apply it to one concave mirror numerical. (PYQ 2024)

Q6. Derive or explain the mirror formula $1/f = 1/v + 1/u$. Also derive the magnification formula $m = -v/u$. State how magnification helps to identify image nature and size. (PYQ 2020)

Q7. A 5 cm tall object is placed 30 cm in front of a concave mirror of focal length 15 cm. Calculate image distance, magnification, image height and nature of image. Show all steps using sign convention. (PYQ 2019)

Q8. A convex mirror has focal length 20 cm. An object is placed 30 cm in front of it. Calculate image distance and magnification. State the position and nature of image. (PYQ 2022)

Q9. Describe refraction of light. State the laws of refraction. Explain why light bends towards the normal in denser medium and away from the normal in rarer medium. Include a labelled diagram. (PYQ 2020)

Q10. Explain refraction through a rectangular glass slab. Draw a labelled diagram showing incident ray, refracted ray, emergent ray, angle of incidence, angle of refraction, angle of emergence and lateral displacement. (PYQ 2019)

Q11. Define refractive index. Explain absolute refractive index and relative refractive index. Calculate refractive index when the speed of light in a medium is 2.25×10^8 m/s. (PYQ 2022)

Q12. Explain the apparent bending of a pencil in water and the apparent raising of a coin placed in water using refraction. Draw diagrams wherever necessary. (PYQ 2024)

Q13. Compare reflection from spherical mirrors and refraction through lenses across five parameters: phenomenon, formula, reference point, image formation, and practical uses. (PYQ 2020)

Q14. Describe convex and concave lenses. Define optical centre, principal axis, principal focus, focal length and aperture. Explain why convex lens is converging and concave lens is diverging. (PYQ 2019)

Q15. Draw ray diagrams for a convex lens for all object positions: at infinity, beyond $2F_1$, at $2F_1$, between F_1 and $2F_1$, at F_1 , and between F_1 and optical centre. Write position, size and nature of image. (PYQ 2022)

Q16. Draw ray diagrams for a concave lens. Explain image formation for an object at infinity and for an object at finite distance. State why the image is always virtual, erect and diminished. (PYQ 2020)

Q17. Explain the lens formula $1/f = 1/v - 1/u$ and magnification formula $m = v/u$. Mention the sign convention for convex and concave lenses. (PYQ 2019)

Q18. A convex lens of focal length 20 cm forms an image of an object placed 30 cm from it. Find image distance, magnification, image height if object height is 4 cm, and nature of image. (PYQ 2024)

Q19. A concave lens of focal length 15 cm forms an image at 10 cm from the lens on the same side as the object. Find object distance and magnification. State the nature of image. (PYQ 2022)

Q20. Define power of a lens. Derive its relation with focal length in metre. Calculate the power of lenses of focal lengths +25 cm, -20 cm and +50 cm. (PYQ 2020)

Q21. Compare mirrors and lenses using a table. Include: phenomenon involved, types, formula, sign of focal length, nature of image, and common uses. (PYQ 2019)

Q22. A student is given a concave mirror, convex mirror, convex lens and concave lens. Explain how the student can identify each one using touch, image formation and behaviour of parallel rays. (PYQ 2024)

Q23. Case Study: A student performs an experiment to find the focal length of a convex lens by obtaining a sharp image of a distant object on a screen. Answer: principle used, nature of image, precautions, reason for using distant object, and formula for power. (PYQ 2022 case-based)

Q24. Case Study: A vehicle uses a convex mirror as rear-view mirror. Answer: type of image formed, reason for wide field of view, limitation of size estimation, sign of focal length and one safety advantage. (PYQ 2020 case-based)

Q25. Write a comprehensive note on Light - Reflection and Refraction covering: laws of reflection, spherical mirrors, mirror formula, refraction, glass slab, spherical lenses, lens formula, magnification and power of lens. (PYQ 2022 comprehensive)

Q26. Create a formula-based answer sheet for this chapter. Include mirror formula, lens formula, magnification of mirror, magnification of lens, relation between R and f, refractive index formula, power of lens formula, SI units, sign convention and one numerical example for each formula. (PYQ 2024 comprehensive)